作业3：（第12次上课时交）

P282	2.3.2	2.3.4	2.3.6	2.3.7	2.3.8	2.3.10
P285	2.3.15	2.3.16(1)		2.3.17		
P247	2.2.3	2.2.4	2.2.7			
P249	2.2.12	2.2.13				

2.3.2

(1) $iR = V_S$ $V_S - V_O = 2iR$

$\therefore V_O = -V_S$

(2) $iR + iR = 0$ $i = 0$

$\therefore V_O = V_S$

(3) $iR = 0$ $\therefore V_O = V_S$

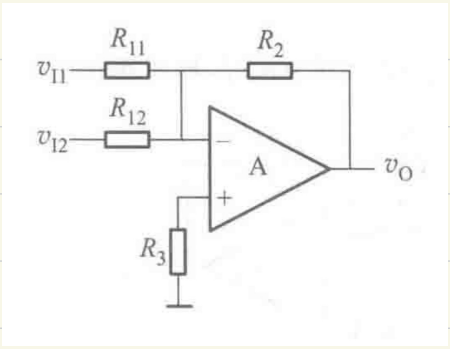
$i = 0$

(4) $iR = V_S$ $2iR = V_S - V_O$

$V_O = -V_S$

2.3.4

(1)

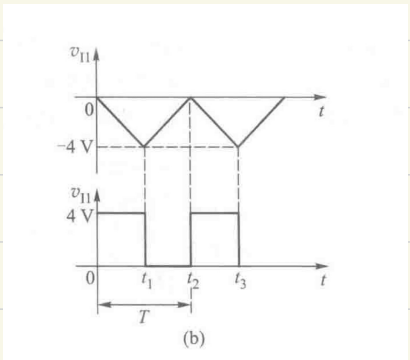


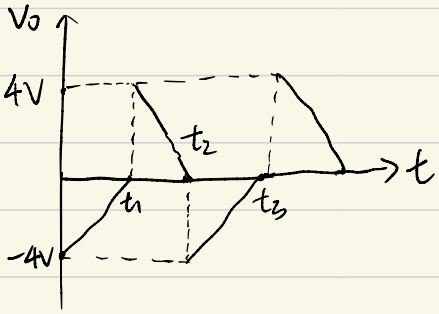
叠加定理

$-\frac{V_{O1}}{R} = \frac{\frac{V_{I1}}{2}}{R//R}$ $-\frac{V_{O2}}{R} = \frac{\frac{V_{I2}}{2}}{R//R}$

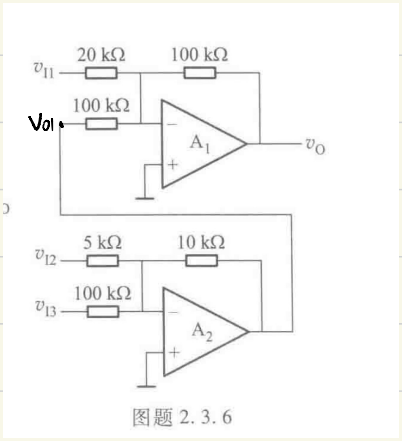
$\therefore V_O = V_{O1} + V_{O2}$

$= -V_{I1} - V_{I2}$





2.3.6



图题 2.3.6

$$V_{O1}=-\frac{100k\Omega}{100k\Omega+5k\Omega}\cdot V_{I2}\cdot\frac{10k\Omega}{100k\Omega//15k\Omega}-\frac{5}{100k\Omega+5k\Omega}\cdot V_{I3}\cdot\frac{10k\Omega}{100//15k\Omega}$$

$$=-2V_{I2}-0.1V_{I3}$$

同理

$$V_O=\frac{-100}{20+100}\cdot V_{I1}\frac{100}{100//20}-\frac{20}{20+100}V_{O1}\cdot\frac{100}{100//20}$$

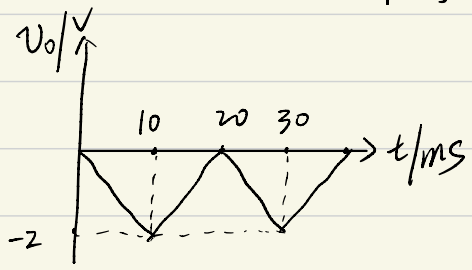
$$=-5V_{I1}-V_{O1}$$

$$\therefore V_O=-5V_{I1}+2V_{I2}+0.1V_{I3}$$

2.3.7 $i=\frac{dV_C}{dt}C$

$$V_I=iR \qquad V_I-iR-V_C=V_O$$

$$\therefore V_O=-V_C=-\frac{1}{RC}\int_0^t V_I\,dt=-100\int_0^t V_I\,dt$$



2.3.8 (1) $V_{O1}=-\frac{R_4}{R_1}V_{S1}+\frac{R_3}{R_2+R_3}V_{S2}\cdot\frac{R_1+R_4}{R_1}$

$$i_C=\frac{V_{S3}}{R_6}+\frac{V_{O1}}{R_5}$$

$$V_O=-\frac{1}{C}\int_0^t i_C dt=-\frac{1}{C}\int_0^t\left(\frac{V_{S3}}{R_6}-\frac{R_4}{R_1R_5}V_{S1}+\frac{R_3(R_1+R_4)}{R_1(R_2+R_3)R_5}V_{S2}\right)dt$$

(2) R时 $V_{O1}=-V_{S1}+V_{S2}$

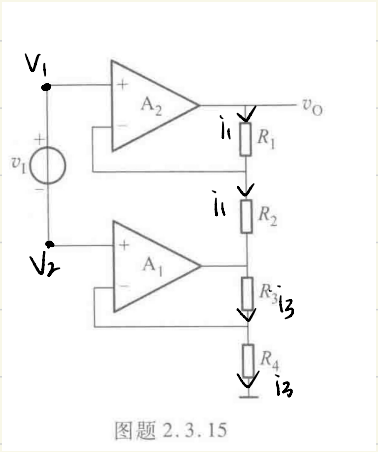
$$V_O=-\frac{1}{RC}\int_0^t(V_{S3}-V_{S1}+V_{S2})dt$$

2.3.10 (1) $\frac{V_1-V_2}{R_1}=i_1$ $V_C=V_1-i_1R_1=6V$

$V_b=0$ $V_e=-0.7V$

(2) $i_1=\frac{V_1-V_2}{R_1}=\beta i_b=\frac{0.2V}{R_2}\beta$ $\beta=50$

2.3.15



图题 2.3.15

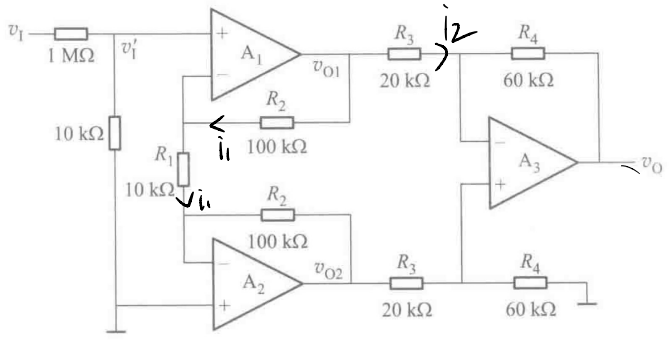
$i_1=\frac{V_O-V_1}{R_1}$ $i_3=\frac{V_1-i_1R-V_2}{R}=\frac{V_1-V_2}{R}-i_1$

$V_O=i_1R\cdot 2+i_3R\cdot 2$

$=2(V_1-V_2)=2V_I$

$\therefore A_U=\frac{V_O}{V_I}=2$

2.3.16 (1)



图题 2.3.16

$V_I'=\frac{10k}{1M+10K}V_I$

$V_{O2}=V_I'-\frac{V_I'}{R_1}(R_1+R_2)$

$=-\frac{R_2}{R_1}V_I'$

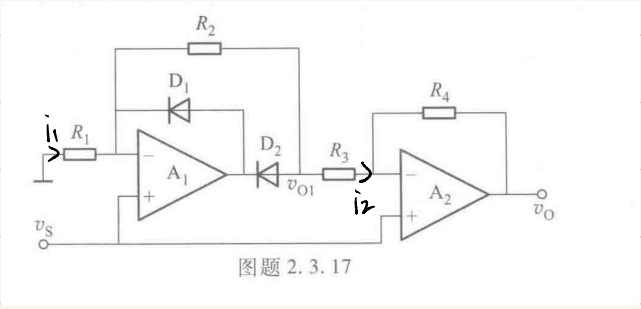
$V_{O1}=V_I'+R_2\frac{V_I'}{R_1}$

$\frac{V_{O1}-V_{O2}}{R_3}+\frac{V_{O2}}{R_3+R_4}=i_2$

$V_O=V_{O1}-i_2(R_3+R_4)=\frac{-63}{101}V_I$

$\therefore A_{vd}=\frac{\Delta V_O}{\Delta V_i}=\frac{V_O}{V_I}=-\frac{63}{101}\approx -0.62$

2.3.17



图题 2.3.17

① $v_S>0$ 时

$i_1=-\frac{v_S}{R_1}$ 二极管导通

$V_{O1}=0-i_1R_1$

$=V_S$

$i_2=\frac{V_S-V_S}{R_3}$ $V_O=V_{O1}-i_2(R_3+R_4)=V_S$

② $V_S<0$ $i_1=-\frac{V_S}{R_1}$ 二极管截止

$V_{O1}=0-i_1(R_2+R_1)=2V_S$ $i_2=\frac{2V_S-V_S}{R_3}$

$V_O=V_{O1}-i_2(R_3+R_4)=-V_S$

∴综上①②，$V_O=|V_S|$

2.2.3 (a) 电压并联负反馈 稳定 V_O

(b) 电流串联负反馈 稳定 I_O

(c) 电流并联负反馈 稳定 I_O

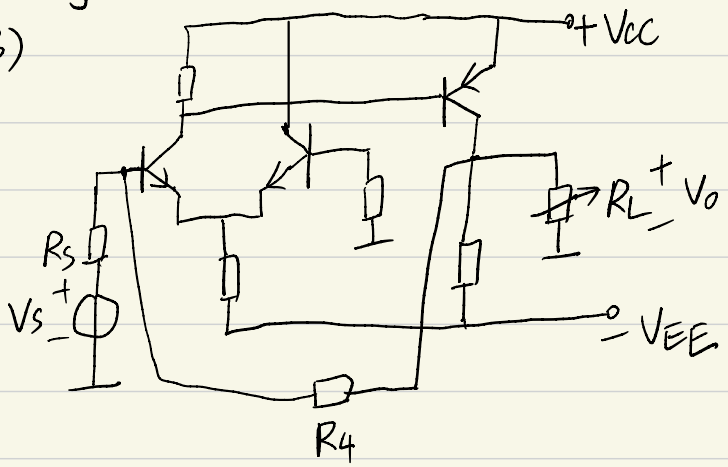
(d) 电压串联负反馈 稳定 V_O

(e) 电压并联负反馈 稳定 V_O

(f) 电压串联负反馈 稳定 V_O

(g) 电压串联负反馈 稳定 V_O

(3)



2.2.4 (a) $R_{if} = 0 + R_1 = R_1$

$R_{of} = 0$

$$i = \frac{V_S}{R_1} = \frac{-V_O}{R_4 + R_2 // R_3} \cdot \frac{R_3}{R_2 + R_3}$$

$$\therefore A_{vf} = \frac{V_O}{V_S} = \frac{-[R_4(R_2+R_3) + R_2R_3]}{R_1R_3}$$

(b) $R_{if} = +\infty$ $R_{of} = +\infty$

$$A_{vf} = \frac{V_O}{V_S} = \frac{R_L}{R_1}$$

(c) $R_{if} = 0 + R_1 = R_1$

$R_{of} = +\infty$

$$A_{vf} = \frac{V_O}{V_S} = -\frac{R_L}{R_1}$$

(d) $R_{if} = +\infty$, $R_{of} = 0$

$$A_{vf} = \frac{(R_1+R_2)R_3 + (R_1+R_2+R_3)R_4}{R_3R_1}$$

(e) $R_{if} = 0 + R_S = R_S$

$R_{of} = 0$

$$i = -\frac{V_O}{R_1} = \frac{V_S}{R_S}$$

$$\therefore A_{vf} = \frac{V_O}{V_S} = -\frac{R_1}{R_S}$$

(f) $R_{if} = \infty$

$R_{of} = 0$

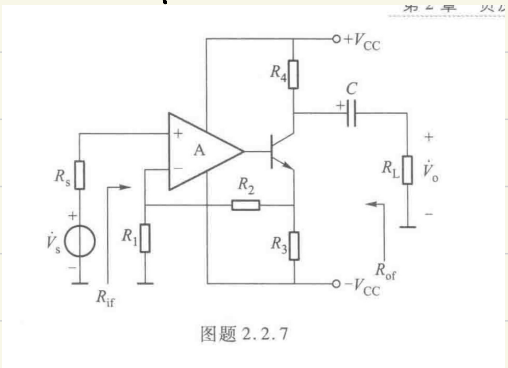
$$V_f = \frac{V_O}{R_3 + R_4} \cdot R_3 \qquad A_{vf} = \frac{1}{F_V} = \frac{R_3 + R_4}{R_3}$$

(g) $R_{if} = \infty$

$R_{of} = 0$

$$A_{vf} = \frac{1}{F_V} = \frac{R_1 + R_2}{R_1}$$

2.2.7



图题 2.2.7

(1) 电流串联负反馈

(2) $$F = \frac{V_f}{I_O} = \frac{-I_O[(R_1+R_2)//R_3]\frac{R_1}{R_1+R_2}}{I_O} = \frac{-R_1R_3}{R_1+R_2+R_3}$$

$$A_{vf} = \frac{V_O}{V_S} = \frac{-(R_1+R_2+R_3)}{R_1R_3}(R_4 // R_L)$$

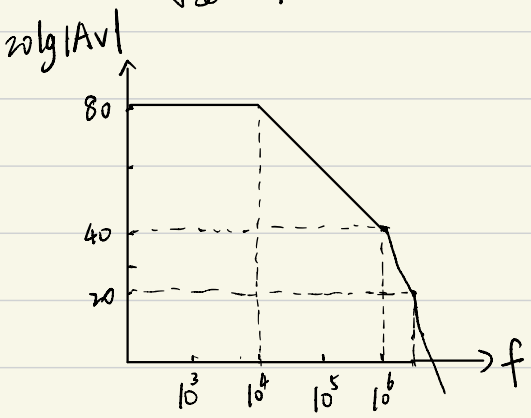
(3) $R_{if} = \infty + R_S = \infty$

$R_{of} = \infty // R_4 = R_4$

2-2.12

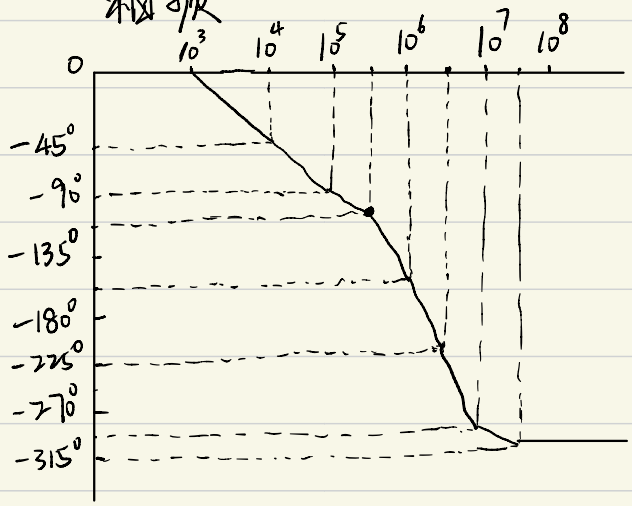
$$\dot{A}_{vd}=\frac{\dot{A}_{vdm}}{\left(1+j\frac{f}{f_{p1}}\right)\left(1+j\frac{f}{f_{p2}}\right)\left(1+j\frac{f}{f_{p3}}\right)}$$

(1) 对数幅频



151-5*90

相频



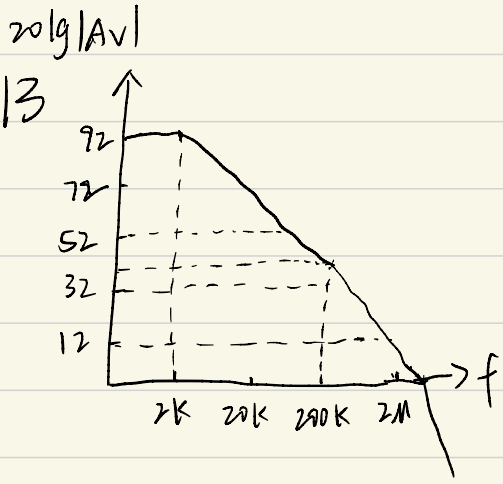
(2) $\varphi=-180^\circ$ 对应 $f=3MHz$ 此时 $20\lg|Av|=25dB$

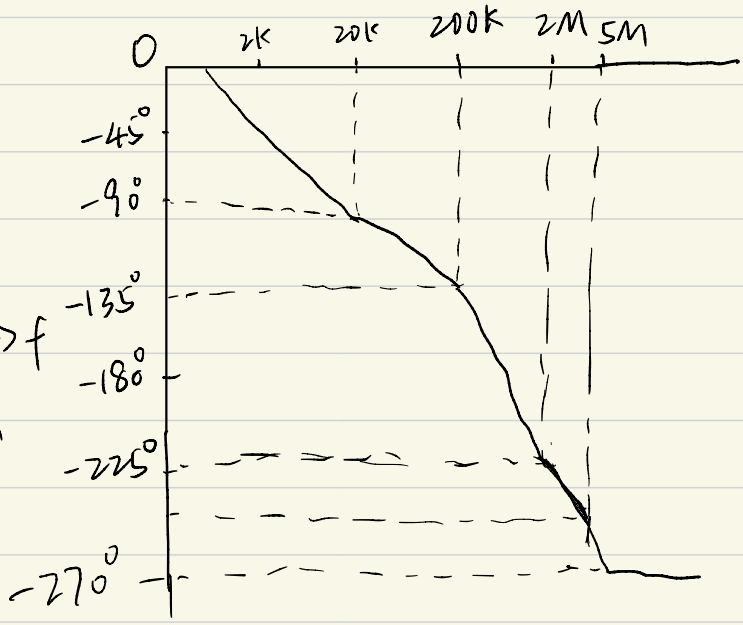
减小至25dB时产生

(3) $\varphi_m=180^\circ-|\varphi_a|=45^\circ$ $\varphi_a=-135^\circ$

则此时 $f=750kHz$，$20\lg|Av|=50dB$

2-2.13





(1) $20\lg|Avf| = 52$

$\varphi_a = -135^\circ \quad \therefore \varphi_m = 180^\circ - |\varphi_a| = 45^\circ$

(2) $20\lg|Avf| = 40$

$\varphi_a = -145^\circ \quad \therefore \varphi_m = 180^\circ - |\varphi_a| = 35^\circ$